



LedgerAI – Document Intelligence Platform

Live: <https://ledger-ai-y8ff.onrender.com>

GitHub: <https://github.com/Srishtikumari510/ledger-ai>

AI Engineer Internship
Technical Case Study

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The Problem

- Manual extraction from financial PDFs is **slow and repetitive**
- Human transcription introduces **data-entry and calculation errors**
- Different layouts make rule-based extraction **fragile**
- Scanned documents require OCR and are vulnerable to **OCR noise**
- Financial results must be **auditable, explainable, and verifiable**



The Solution

Native PDFs use text extraction first; scanned documents use smart OCR fallback

Gemini converts document content into type-specific structured JSON

Deterministic rules independently verify financial relationships.

**Upload → Validate → OCR → AI Extract → Structured JSON →
Financial Validation → Store → Dashboard + REST API.**

Architecture

LedgerAI - Architecture

DOCUMENT INTELLIGENCE PIPELINE

LAYER 1 - CLIENTS

Landing page (HTML/CSS/JS) Dashboard (Jinja2) Swagger / OpenAPI REST clients (curl / Postman)

LAYER 2 - API

FastAPI · POST /api/v1/documents/process · GET /api/v1/documents/{name} · GET /api/v1/documents · GET /api/v1/health

LAYER 3 - ORCHESTRATION

document_service.py — coordinates validation → OCR → extraction → financial validation → persistence

STEP 1

File Validation

- Type (PDF/JPG/PNG)
 - Size, readability
 - Page count ≤ 3
- PASS / FAILED

STEP 2

OCR / Text Extract

- pdftplumber (native)
 - Tesseract (fallback)
 - Poppler (PDF → img)
- Per-page text

STEP 3

LLM Extraction

- Google Gemini
 - Strict JSON schema
 - Page-level evidence
- Structured fields

STEP 4

Financial Validation

- Deterministic rules
 - Formula + operands
 - Variance + status
- PASS / FAIL / N/A

LAYER 5 - PERSISTENCE

Repository · SQLAlchemy ORM

SQLite (dev) · PostgreSQL-ready (prod)

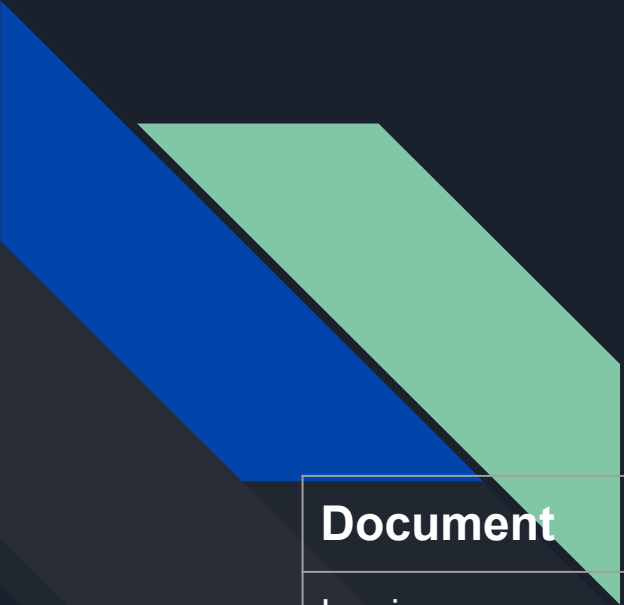
LAYER 6 - OUTPUT

Consistent structured JSON

document_name · file_validation · extracted_data · validation · metadata

Runtime: Python 3.12 · Uvicorn · Docker · Render
Deployment: <https://ledger-ai-y8ff.onrender.com>

LedgerAI · 2026



Documents Supported

Document	Minimum fields
Invoice	Invoice number, date, vendor, subtotal, tax, total
Balance Sheet	Assets, liabilities, equity, reporting period
Profit & Loss	Revenue, expenses, net profit, reporting period
Cash Flow	Operating, investing, financing, net cash flow



Extraction Approach

Native-first: pdfp1umber extracts embedded PDF text

Smart fallback: OCR activates when extracted text is unreliable

Scanned PDFs: Tesseract 5 + Poppler

LLM extraction: Gemini with strict per-type JSON schemas

Evidence: every field includes page_number + source_text

Missing values: null — never invented



Financial Validation

Deterministic rules: AI does not decide whether calculations pass
Every check returns formula + operands + calculated + reported + variance + status
Handles bracketed negatives: (1234) → -1234
Comparative statements use **per-period validation**
Tolerance: $\max(0.01, 0.001 \times |\text{reported}|)$

Example checks

- **Invoice:** $\text{subtotal} + \text{tax} \approx \text{total}$
- **Balance Sheet:** $\text{assets} \approx \text{liabilities} + \text{equity}$
- **P&L:** $\text{revenue} - \text{expenses} \approx \text{net_profit}$
- **Cash Flow:** $\text{opening} + \text{net_change} \approx \text{closing}$

Working Demo

LedgerAI

Solutions Platform How It Works Case Studies Technology API Docs

Launch Dashboard →

SOLUTIONS

Built for Real-World Enterprise Challenges

Six core capabilities that map directly to how finance and operations teams work.

01

Intelligent Document Processing

Extract structured information from invoices, balance sheets, P&L, and cash flow statements using AI with page-level evidence.



02

AI Validation

Every extracted value is checked against financial relationships and business rules — with traceable PASS / FAIL / NOT_APPLICABLE outcomes.



03

Financial Intelligence

Reconcile subtotals, taxes, assets, liabilities, cash flows, and profit line items automatically across comparative periods.



04

AI APIs

Expose every capability through consistent, documented REST APIs. Multipart upload, GET-by-name, list, and health endpoints out of the box.



05

Analytics

Turn extracted data into structured JSON, searchable dashboards, and machine-readable outputs ready for downstream systems.



06

Automation

Remove manual data entry from document-heavy workflows. Upload once, get validated structured output for every page.



LedgerAI

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Launch Dashboard →

ENTERPRISE DOCUMENT INTELLIGENCE

Turn Complex Documents Into Structured Intelligence

Build, validate, and operationalize AI systems that transform enterprise documents into measurable business outcomes — with accuracy you can audit.

Explore the Platform

Launch Dashboard →

Audit-grade validation · Page-level evidence · Structured JSON APIs

LedgerAI

Solutions Platform How It Works Case Studies Technology API Docs

Launch Dashboard →

99.2%

EXTRACTION ACCURACY

10K+

DOCUMENTS PROCESSED

60%

FASTER PROCESSING

24/7

AUTOMATED VALIDATION

INTELLIGENCE THAT CREATES VALUE

From Raw Data to Reliable Intelligence

LedgerAI combines document AI, deterministic validation rules, structured APIs, and audit-grade evidence to turn messy PDFs, scans, and images into production-grade data. Every extracted value is traceable. Every financial relationship is checked. Every result is served through clean REST endpoints.

01 DATA → 02 AI EXTRACTION → 03 VALIDATION → 04 STRUCTURED OUTPUT
→ 05 BUSINESS ACTION

Working Demo

LedgerAI

SolutionsPlatformHow It WorksCase StudiesTechnologyAPI DocsLaunch Dashboard --

THE PLATFORM

One Platform. Complete Intelligence Workflow.

Upload, extract, validate, and retrieve — all through one unified service.

LedgerAI - Processing

Document	Invoice_2026.pdf
Status	✓ Processed
Confidence	97.8%
Validation	✓ Passed
Fields Extracted	42
OCR Used	Yes

PROCESSING PIPELINE

Validation

100%

response.json

200 OK

```
{
  "document_name": "Invoice_2026.pdf",
  "document_type": "invoice",
  "processing_status": "PASS",
  "extracted_data": {
    "invoice_number": "INV-2026-0042",
    "vendor_name": "Acme Supplies",
    "currency": "USD",
    "subtotal": 2530.00,
    "tax_amount": 253.00,
    "total_amount": 2783.00,
    "line_items": [
      {
        "description": "Office Chair",
        "quantity": 5,
        "amount": 600.00
      }
    ]
  },
  "validation": {
    "overall_status": "PASS",
    "checks": [
      {
        "name": "invoice_total_check",
        "formula": "subtotal + tax - discount",
        "calculated_value": 2783.00,
```

LedgerAI

SolutionsPlatformHow It WorksCase StudiesTechnologyAPI DocsLaunch Dashboard --

HOW IT WORKS

Four Steps From Upload to Insight

01

Upload
Drop a PDF, JPG, or PNG through the dashboard or POST it to the REST API.

02

Extract
OCR + LLM pipelines identify every meaningful field and table with page-level evidence.

03

Validate
Financial relationships and business rules verify extracted values with explicit formulas.

04

Deliver
Clean structured JSON is returned, persisted, and retrievable by document name.

CASE STUDIES

Designed for Production

FINANCIAL SERVICES

SHARED SERVICES

AUDIT & COMPLIANCE

LedgerAI

SolutionsPlatformHow It WorksCase StudiesTechnologyAPI DocsLaunch Dashboard --

CASE STUDIES

Designed for Production

FINANCIAL SERVICES

Automating Complex Financial Document Processing
Balance sheets and cash flow statements parsed into structured JSON with period-aware validation.

95%+ automation 60% faster processing
[View example --](#)

SHARED SERVICES

Invoice Data Extraction at Scale
Line-item reconciliation across quantity x unit price x amount, with tax and discount checks.

99% field accuracy Zero manual entry
[View example --](#)

AUDIT & COMPLIANCE

Traceable Validation for Regulatory Review
Every check returns formula, operands, calculated value, reported value, and variance.

100% traceable Audit-ready outputs
[View example --](#)

Working Demo

LedgerAI

Launch Dashboard →

Ready to Turn Data Into Intelligence?

Build reliable AI workflows that extract, validate, and operationalize information at enterprise scale.

[Launch Dashboard](#)[Explore the API](#)

LedgerAI

Enterprise document intelligence for financial services, audit, and shared services teams.

SOLUTIONS

- Document Intelligence
- AI Validation
- Financial Intelligence
- Analytics
- Automation

PLATFORM

- Dashboard
- API Documentation
- Health Check
- How it Works

COMPANY

- About
- Case Studies
- LinkedIn
- FAQ

LedgerAI

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ARCHITECTURE

A Clean, Layered Pipeline

CLIENT / BROWSER

↓

FASTAPI REST API

↓

VALIDATION SERVICE

↓

OCR SERVICE

↓

LLM EXTRACTION

↓

FINANCIAL VALIDATION

↓

REPOSITORY / DB

↓

STRUCTURED JSON

INSIGHTS

Insights & Intelligence

DOCUMENT AI 6 MIN READ

Why page-level evidence matters in document extraction

Confidence without provenance is a liability. How we attach evidence to every extracted field.

[Read more →](#)

VALIDATION 5 MIN READ

Financial validation is not a black box

Every check we run returns a formula, operands, calculated value, reported value, and variance.

[Read more →](#)

ENGINEERING 8 MIN READ

Building OCR + LLM pipelines that survive real documents

Native text first, OCR fallback, smart heuristics, structured LLM prompts — a practical guide.

[Read more →](#)

FAQ

Frequently Asked Questions

What documents can the platform process?

+

How accurate is the AI extraction?

+

How does validation work?

+

Can the platform integrate with existing systems?

+

Is there a REST API?

+

How is sensitive data protected?

+

Working Demo

LedgerAI

Home Dashboard API Docs

Back to Home

Document Dashboard

Upload financial documents for AI-powered extraction & validation.

Process a Document

Document Type

Invoice

File (PDF / JPG / PNG, max 3 pages)

Choose file Consolidated Balance Sheet 2024.pdf

Process Document

Processed Documents

Refresh

DOCUMENT NAME	TYPE	STATUS	PROCESSED AT	
Consolidated Balance Sheet 2019.pdf	balance_sheet	FAILED	11/09/2026, 16:51:09	View →
Consolidated Balance Sheet 2020.pdf	balance_sheet	FAILED	11/09/2026, 16:50:13	View →
Consolidated Balance Sheet 2021.pdf	balance_sheet	FAILED	11/09/2026, 16:48:43	View →
Consolidated Balance Sheet 2023.pdf	balance_sheet	FAILED	11/09/2026, 16:47:40	View →
Consolidated Balance Sheet 2025.pdf	balance_sheet	FAILED	11/09/2026, 16:46:26	View →

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Processed Documents

Refresh

DOCUMENT NAME	TYPE	STATUS	PROCESSED AT	
Consolidated Balance Sheet 2019.pdf	balance_sheet	FAILED	11/09/2026, 16:51:09	View →
Consolidated Balance Sheet 2020.pdf	balance_sheet	FAILED	11/09/2026, 16:50:13	View →
Consolidated Balance Sheet 2021.pdf	balance_sheet	FAILED	11/09/2026, 16:48:43	View →
Consolidated Balance Sheet 2023.pdf	balance_sheet	FAILED	11/09/2026, 16:47:40	View →
Consolidated Balance Sheet 2025.pdf	balance_sheet	FAILED	11/09/2026, 16:46:26	View →
Consolidated Balance Sheet 2026.pdf	balance_sheet	FAILED	11/09/2026, 16:45:08	View →
Consolidated Balance Sheet 2024.pdf	balance_sheet	FAILED	11/09/2026, 16:21:09	View →
Consolidated Profit & Loss 2025.pdf	profit_and_loss	PASS	11/09/2026, 13:42:26	View →
Consolidated Cash Flow Statement 2024.pdf	cash_flow_statement	FAILED	11/09/2026, 13:41:22	View →
Consolidated Cash Flow Statement 2025.pdf	cash_flow_statement	PASS	11/09/2026, 13:36:43	View →
Consolidated Balance Sheet 2022.pdf	balance_sheet	FAILED	11/09/2026, 13:03:15	View →
Consolidated Balance Sheet 2024.pdf	balance_sheet	PASS	11/09/2026, 12:42:38	View →
Document 4.pdf	invoice	PASS	11/09/2026, 11:59:19	View →

LedgerAI

Home Dashboard API Docs

Dashboard

Consolidated Profit & Loss 2025.pdf

Processed document result

Summary

Type: profit_and_loss

Status: PASS

Confidence: N/A

Processed: 2026-09-11T08:12:26.114577+00:00

File Validation

File type	application/pdf
Supported	true
Readable	true
Pages	1
Status	PASS

Extracted Fields

periods (p:1) ["March 31, 2025","March 31, 2024"] currency (c:1) INR entity_name N/A statement_date (s:1) March 31, 2025 revenue (r:1) 470915.93 cost_of_sales N/A gross_profit N/A operating_expenses (o:1) 176605.07 operating_profit N/A tax N/A net_profit (p:1) 70792.25

Working Demo

LedgerAI

Home

Dashboard

API Docs

← Dashboard

Processing Metadata

OCR used	true
LLM used	true
Provider	gemini
Model	gemini-3.6-flash
Processing time	23189 ms

Hide Raw JSON

```
{
  "document_name": "Consolidated Profit & Loss 2025.pdf",
  "document_type": "profit_and_loss",
  "processing_status": "PASS",
  "overall_confidence": null,
  "file_validation": {
    "file_type": "application/pdf",
    "is_supported": true,
    "is_readable": true,
    "page_count": 1,
    "status": "PASS",
    "error_code": null,
    "error_message": null
  },
  "extracted_data": {
    "periods": {
      "value": [
        "March 31, 2025",
        "March 31, 2024"
      ]
    }
  }
}
```

Document Intelligence API 1.0.0 QAS 1.1

OpenAPI spec

AI-powered document extraction and validation API

documents

- GET /api/v1/health Health
- POST /api/v1/documents/process Process
- GET /api/v1/documents List All
- GET /api/v1/documents/{document_name} Get By Name

default

- GET / Landing
- GET /dashboard Dashboard
- GET /result/{document_name} Result Page

Schemas

- Body_process_api_v1_documents_process_post > Expand all object
- DocumentListItem > Expand all object

default

- GET / Landing
- GET /dashboard Dashboard
- GET /result/{document_name} Result Page

Schemas

- Body_process_api_v1_documents_process_post > Expand all object
- DocumentListItem > Expand all object
- DocumentResponse > Expand all object
- FileValidation > Expand all object
- HTTPValidationError > Expand all object
- ProcessingMetadata > Expand all object
- ValidationCheck > Expand all object
- ValidationError > Expand all object
- ValidationResult > Expand all object



REST API

POST /api/v1/documents

GET /api/v1/documents

GET /api/v1/documents/{id}

GET /api/v1/health

- Consistent structured JSON responses
- HTTP status codes for success and failure cases
- Swagger/OpenAPI documentation available at /docs
- API base: /api/v1



Tech Stack

- **Backend:** FastAPI · Pydantic v2 · SQLAlchemy · Uvicorn · SQLite
- **OCR:** Tesseract 5 · Poppler · pdfplumber · pdf2image
- **LLM:** Google Gemini · structured JSON extraction
- **Frontend:** Jinja2 · HTML · CSS · JavaScript
- **Deployment:** Docker on Render
- **Testing:** pytest — **10/10 passing**

Known Limitations → Production Path

Free Tier	Production
Synchronous processing	Async job queue
SQLite	PostgreSQL
Ephemeral storage	Persistent object storage
Basic extraction	Confidence scoring + human review
No authentication	OAuth/JWT + role-based access

AI Usage + Closing

- **LLM in product: Gemini for structured financial extraction**
- **AI-assisted development: Claude / ChatGPT / Gemini for scaffolding, prompts & debugging**
- **No hardcoded document outputs**

AI is used where it provides the most value: understanding variable document content and producing structured fields. Critical financial calculations remain deterministic, ensuring the system does not blindly trust generated values.

Thank you